Tracking **stock, assets, and resource**

1. Two Types of Resource Tracking

Your system should separate **consumable stock** and **fixed assets**.

Consumable Stock

Items that get used up.

Examples:

- chalk
- exam papers
- printer ink
- cleaning supplies
- cafeteria food ingredients

These reduce in quantity when issued.

Fixed Assets

Items that remain in use long-term.

Examples:

- computers
- projectors
- desks
- laboratory equipment
- buses
- kitchen appliances

These are **assigned rather than consumed**.

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2. Central Store Architecture

The best architecture is **one central inventory module** that serves all other modules.

Structure:

```
Inventory & Assets Module
   ├ Stock Management
   ├ Asset Management
   ├ Store Requests
   ├ Procurement
   ├ Suppliers
   ├ Stock Allocation
```

Other modules **request resources from the store**.

Example modules requesting stock:

- Academics
- Exams
- Library
- Cafeteria
- Transport
- Maintenance
- Administration

---

3. Stock Tracking Workflow (Real School Scenario)

Example: Teacher needs **50 exam papers**.

Step 1 – Request

Teacher or department submits request.

```
Module: Exams
Item: A4 Paper
Quantity: 50
Reason: Midterm Exams
```

Step 2 – Store Manager Review

Store manager checks:

- available stock
- request reason
- department allocation

Actions:

```
Approve
Reject
Modify quantity
```

Step 3 – Stock Issued

System deducts inventory.

Example:

```
Stock before: 500
Issued: 50
Stock remaining: 450
```

Record saved.

---

4. Asset Allocation Workflow

Assets are **assigned**, not deducted.

Example:

```
Asset: Laptop
Serial: LTP-2034
Assigned to: Teacher John
Department: Science
```

System tracks:

- who has the asset
- location
- condition
- maintenance history

---

5. Store Request Module

All modules should send requests through **Store Requests**.

Structure:

```
Store Requests
   ├ Pending
   ├ Approved
   ├ Issued
   ├ Rejected
```

Example request:

```
Department: Cafeteria
Item: Rice
Quantity: 20kg
Purpose: Weekly meals
```

---

6. Roles for Inventory System

Store Manager

Main controller.

Permissions:

- manage stock
- approve requests
- issue items
- record deliveries

Procurement Officer

Handles purchasing.

Permissions:

- create purchase orders
- manage suppliers
- approve procurement

Department Heads

Can request items.

Example:

Department	Request Role
Academics	Head of Academics
Cafeteria	Kitchen Manager
Transport	Transport Manager
Maintenance	Facilities Manager

Finance / Bursar

Monitors budget.

Permissions:

- approve high-cost purchases
- view spending reports

Tenant Admin

Full control.

---

7. Item Categories

Inventory should support categories.

Examples:

```
Stationery
Cleaning Supplies
Lab Materials
Food Supplies
Maintenance Tools
Sports Equipment
IT Equipment
```

```
Furniture
```

This helps with budgeting.

---

8. Stock Item Structure

Each stock item should include:

```
Item Name
Category
Unit
Quantity Available
Minimum Stock Level
Supplier
Unit Cost
Storage Location
```

Example:

```
Item: Chalk
Category: Stationery
Unit: Box
Quantity: 120
Minimum Level: 20
```

---

9. Automatic Stock Alerts

System should trigger alerts when stock is low.

Example:

```
Alert: Chalk stock below minimum level.
```

Send to:

- store manager
- procurement officer

---

10. Department Budget Tracking

Each department can have a **budget allocation**.

Example:

```
Department Budget

Academics: $20,000
Cafeteria: $50,000
Transport: $30,000
Maintenance: $10,000
```

When items are issued, the cost is recorded.

Example:

```
50 Exam Papers = $5
```

Budget reduces accordingly.

---

11. Asset Tracking Fields

Asset records should include:

```
Asset Name
Asset Code
Serial Number
Purchase Date
Purchase Cost
Supplier
Assigned Department
Assigned User
Condition
Warranty Expiry
Maintenance Schedule
```

Example:

```
Asset: Projector
Code: PRJ-104
Department: ICT Lab
Condition: Good
```

---

12. Asset Movement Tracking

Assets may move across departments.

Example:

```
Projector moved from Science Lab → Hall
```

System records movement history.

---

13. Maintenance Tracking

Assets need maintenance.

Example:

```
School Bus Service
Next Service Date: April 10
```

Alert should be triggered.

---

14. Inventory Dashboard

Store manager dashboard example:

```
Inventory Dashboard

Total Stock Items: 450
Low Stock Alerts: 12
Pending Requests: 8
Assets Assigned: 320
Assets Under Maintenance: 6
```

---

15. Department Allocation Tracking

System should track:

```
Department → Items Issued
```

Example:

```
Academics Department

Chalk: 30 boxes
A4 Paper: 500 sheets
Markers: 20
```

This helps identify **resource consumption patterns**.

---

16. Audit Trail

Every inventory action must be logged.

Example:

```
March 12
Store Manager issued 20kg rice to Cafeteria
```

Audit fields:

```
action
user
date
quantity
department
```

---

17. Procurement Workflow

If stock runs low:

```
Stock < minimum level
```

System suggests purchase.

Workflow:

```
Store Manager → Request Purchase
Procurement → Create Purchase Order
Finance → Approve Budget
Supplier → Deliver Items
Store → Record Stock
```

---

18. Inventory Reports

Useful reports:

Stock Levels

```
All items and quantities
```

Department Usage

```
Academics used $3,000 this month
```

Asset Register

```
List of all school assets
```

Procurement History

```
Supplier purchases
```

---

19. Integration with Other Modules

Inventory interacts with:

Module	Interaction
Cafeteria	food stock
Exams	exam materials
Library	books inventory
Transport	vehicle assets
Maintenance	repair tools
ICT	computers

---

20. Final System Flow

```
Departments
     ↓
Submit Store Request
     ↓
Store Manager Review
     ↓
Approve / Reject
     ↓
Stock Issued
     ↓
Inventory Updated
     ↓
Budget Updated
```

21. Recommended Database Tables

```
inventory_items
inventory_transactions
store_requests
asset_registry
asset_assignments
suppliers
purchase_orders
department_budgets
```

managing assets, investments, and financial resources

1. What Schools Actually Manage Financially

A tenant (school) typically manages four major financial areas:

1 Operational Finance

Daily school money.

Examples:

- school fees
- salaries
- utility payments
- cafeteria expenses
- maintenance

Handled in:

```
Finance Module
```

---

2 Physical Assets

Things the school owns.

Examples:

- buildings
- land
- buses
- furniture
- computers
- laboratory equipment

Handled in:

```
Asset Management
```

---

3 Investments

Money invested for growth or reserves.

Examples:

- fixed deposits
- treasury bonds
- property investments
- school development funds
- endowment funds

Handled in:

```
Investments Module (Finance submodule)
```

---

4 Development Projects

Long-term capital projects.

Examples:

- new classroom block
- sports field construction
- school bus purchase
- ICT lab upgrade

Handled in:

```
Projects / Capital Development
```

---

2. Where This Lives in Your SMS

Inside the tenant system:

```
Finance Module
   ├ Fees & Billing
   ├ Expenses
   ├ Payroll
   ├ Budgeting
   ├ Investments
   ├ Asset Register
   ├ Capital Projects
   └ Financial Reports
```

---

3. Asset Register (Very Important)

Every school should maintain a **full asset register**.

Example:

Asset	Category	Location	Value
School Bus	Transport	Garage	$40,000
Classroom Block A	Building	Campus	$200,000
Laptop Lab	ICT	Lab 1	$15,000

Fields to store:

```
Asset ID
Asset Name
Category
Purchase Date
Purchase Cost
Current Value
Location
Assigned Department
Maintenance Schedule
Depreciation Rate
```

---

4. Asset Depreciation

Assets lose value over time.

Example:

```
Laptop purchase: $1000
Depreciation: 20% yearly
```

Year values:

```
Year 1 → $800
Year 2 → $640
Year 3 → $512
```

System should calculate **automatic depreciation**.

Useful for:

- financial reporting
- insurance
- budgeting

---

5. Investment Management

Some schools invest surplus funds.

Examples:

Investment	Type
Treasury Bond	Government security
Fixed Deposit	Bank
Real Estate	Property
School Development Fund	Internal fund

Investment record fields:

```
Investment Name
Type
Amount Invested
Start Date
Maturity Date
Interest Rate
```

```
Expected Return
Status
```

Example:

```
Investment: Government Treasury Bond
Amount: $50,000
Interest: 12%
Maturity: 3 years
```

---

6. Investment Tracking

The system should track:

```
Principal Amount
Interest Earned
Maturity Date
Withdrawals
Returns
```

Example:

```
Principal: $50,000
Interest Earned: $6,000
Total Value: $56,000
```

---

7. Capital Projects

Schools often run development projects.

Example:

```
Project: New Science Lab
Budget: $120,000
Spent: $45,000
Remaining: $75,000
```

Fields:

```
Project Name
Budget
Start Date
End Date
Project Manager
Expenses
Completion Status
```

---

8. Budget Management

Each tenant should set **annual budgets**.

Example:

```
Annual Budget 2026
```

Department	Budget
Academics	$80,000
Transport	$30,000
Cafeteria	$50,000
Maintenance	$20,000

System tracks:

```
Budget Allocated
Budget Used
Budget Remaining
```

---

9. Financial Dashboard

Tenant admin / bursar should see a dashboard like:

```
Financial Overview

Total Assets Value: $1,200,000
Active Investments: $150,000
Annual Budget: $500,000
Expenses This Month: $40,000
Revenue This Month: $70,000
```

---

10. Financial Reports

Important reports:

Asset Register Report

```
List of all assets
```

Investment Portfolio

```
All school investments
```

Income vs Expenses

```
Monthly financial performance
```

Budget Utilization

```
Department spending vs budget
```

---

11. Who Manages These in the Tenant

Typical roles:

Bursar / Finance Officer

Main finance controller.

Permissions:

- manage investments
- manage expenses
- track budgets
- manage financial reports

Principal / Headteacher

Oversight.

Permissions:

- approve large expenses
- approve investments
- view reports

Board / Directors

View strategic finances.

Permissions:

- investment reports
- capital projects
- financial summaries

Store Manager

Manages physical assets inventory.

Procurement Officer

Handles purchasing.

---

12. Integration with Other Modules

Finance integrates with many modules.

Module	Integration
Fees	revenue
Payroll	staff salaries
Inventory	stock costs
Transport	vehicle expenses
Maintenance	repair costs
Cafeteria	food expenses

---

13. Asset Lifecycle in the System

Example lifecycle:

```
Purchase Asset
     ↓
Register Asset
     ↓
Assign Department
     ↓
Track Maintenance
     ↓
Depreciate Value
     ↓
Dispose or Replace
```

---

14. Asset Disposal

When assets are sold or scrapped.

Example:

```
Old Computer
Status: Disposed
Sale Value: $100
```

System records disposal.

---

15. Investment Lifecycle

```
Create Investment
     ↓
Track Returns
     ↓
Monitor Maturity
     ↓
Withdraw / Reinvest
```

---

16. Multi-Tenant Consideration (Important)

Each tenant must have **isolated financial records**.

Example:

```
Tenant A → Assets $1M
Tenant B → Assets $400K
```

No tenant should see another tenant's finances.

---

17. Recommended Database Tables

```
assets
asset_assignments
asset_maintenance
investments
investment_returns
financial_budgets
financial_transactions
capital_projects
```

---

18. Alerts and Reminders

The system should generate alerts for:

```
Investment maturity
Asset maintenance due
Budget exceeded
Low department funds
Procurement approvals
```

---

19. Exported Financial Documents

Reports should include **tenant branding**:

- logo
- colors
- school name
- address

Example:

```
Greenfield Academy
Annual Asset Register
```

---

20. Ideal Tenant Finance Structure

Final structure:

```
Finance Module
   ├ Fees
   ├ Expenses
   ├ Payroll
   ├ Budgets
   ├ Investments
   ├ Asset Register
   ├ Capital Projects
   └ Financial Reports
```